

Sea Ice Deep Fusion

Workflow Overview

Combining satellite imagery with ICESat-2 laser altimetry to map Arctic sea ice — thick ice, thin ice, and open water

test mIoU

0.8982

pixel accuracy

95.3%

macro F1

0.9453

1

Thick Ice

2

Thin Ice

3

Water

3-class segmentation · tile-grouped train/test split · focal loss

Why is this tricky?

RGB alone can't reliably tell thin ice from thick — they look almost the same in visible wavelengths

Melt ponds add even more confusion to the colour signal

Thin ice is rare in the dataset but matters a lot for climate work

Test tile (T03CWT) is a completely different region — no overlap with training tiles at all

Our approach

Satellite patches give us spatial texture → U-Net handles this

ICESat-2 height/roughness features → LSTM reads the sequence

We merge both feature sets before the final classification layer

ResNet-18 encoder starts from ImageNet weights (saves a lot of training time), LSTM init from a separate sweep-winner run

SE block to let the network re-weight channels after merging

Satellite patches

128×128 px RGB crops

Tiles: T02CNA, T02CNC → train

T03CWT → test only

Normalised with ImageNet stats

Augmented with flip + rotate

ICESat-2 ATL03

Along-track, ~10 m spacing

8 features per segment:

- h_cor_mean, h_diff
- rel_height_min_elev, height_sd
- pcnth/pcnt/bcnt/brate_mean

5-segment sliding window

Label masks

RGB-coded segmentation maps:

- Red (255,0,0) = thick ice
- Blue (0,0,255) = thin ice / melt pond
- Green (0,255,0) = open water

How we split the data (by tile, not randomly)

Train (~90%)

T02CNA + T02CNC

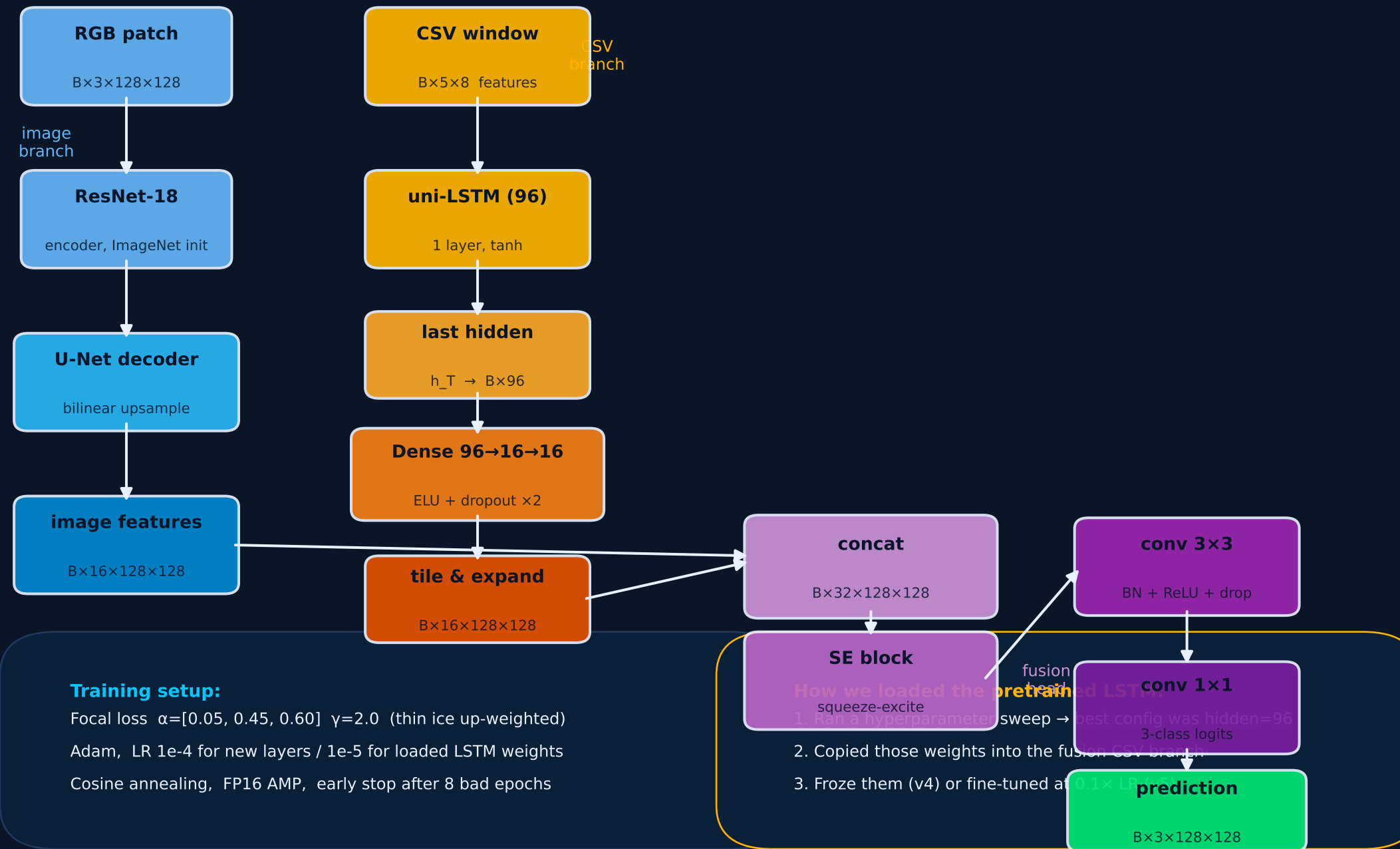
Val (~10%)

random held-out
from train tiles

Test (unseen)

T03CWT only —
never in training

Keeping T03CWT completely out of training means the test numbers reflect real generalisation, not just memorisation.



Step 1 — Match images to CSV rows

Each 128×128 patch is linked to an ATL03 row via tile + beam + row_idx stored in the manifest.

We build 8-feature arrays for every CSV segment and compute z-score stats on train tiles only — test stats never touch the normalisation.

Step 2 — Normalise and cache

CSV features get z-scored using train-tile mean/std, then saved as .npy so we don't recompute each epoch (speeds things up).

Image normalisation uses ImageNet numbers and is applied on-the-fly. Train set also gets random flips and 90° rotations.

Step 3 — Find best LSTM settings first

Before building the fusion model we ran a sweep over hidden size {32,64,96}, LR, dropout, focal γ , α , and window length.

Winner: hidden=96, lr $\approx$ 1e-3, α =[0.05,0.45,0.60], γ =2.0, seq_len=5.

Step 4 — Assemble the fusion model

U-Net (ResNet-18 encoder, ImageNet weights) outputs a 16-channel feature map at full res.

We load the sweep-winner LSTM weights into the CSV branch, then tile the 96-dim hidden state to match the 128×128 spatial size.

Step 5 — Train with two learning rates

Focal loss with α =0.45 on thin ice really helps — without it the model mostly ignores the rare thin-ice class.

Loaded weights get LR=1e-5 (slow adaptation), fresh layers get LR=1e-4. Early stopping kicks in after 8 epochs with no val gain.

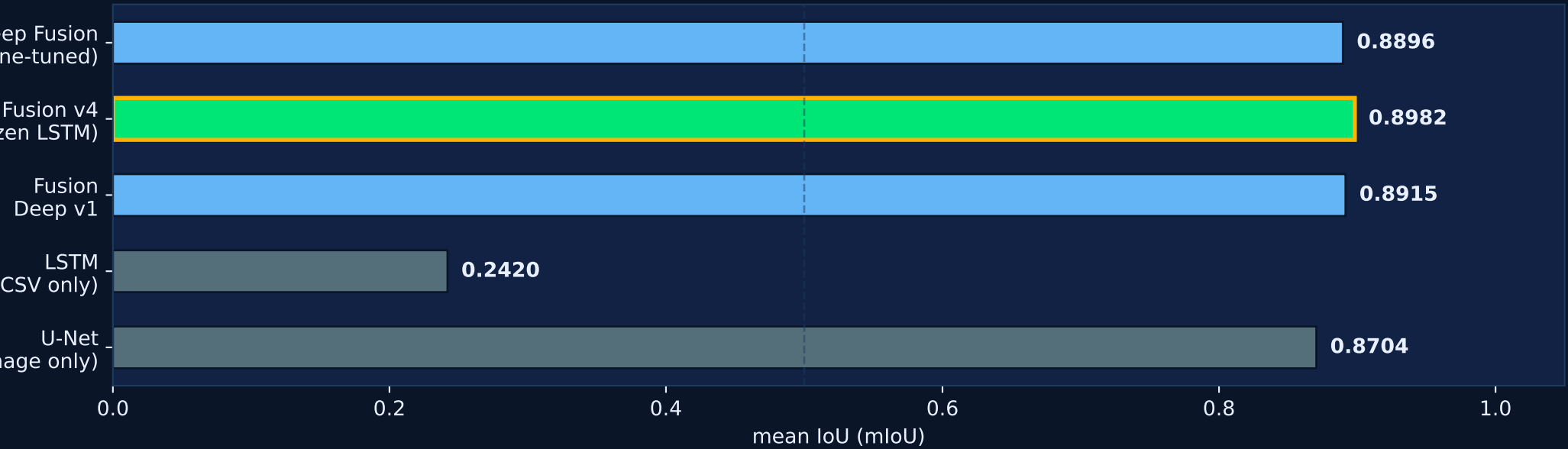
Step 6 — Test on the held-out tile

We load best.pt (epoch with highest val mIoU) and run it on T03CWT — a tile the model has never seen.

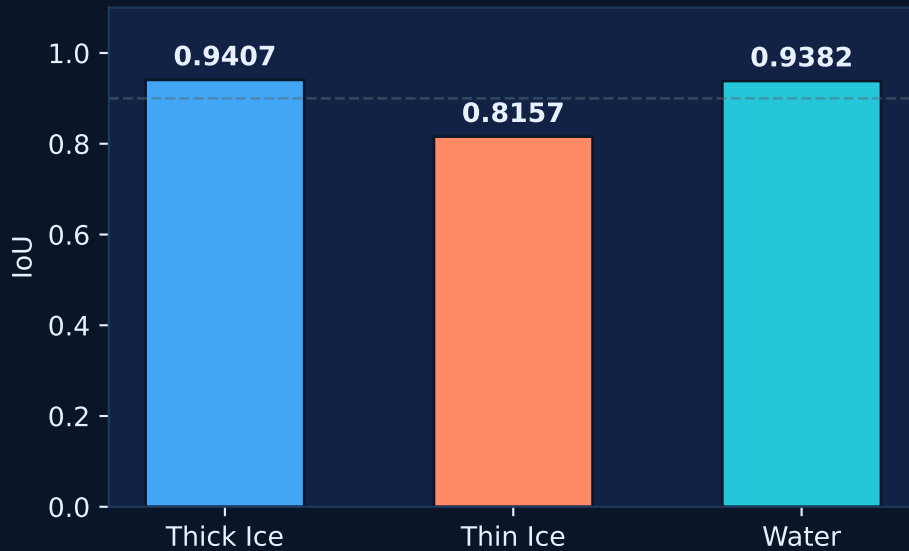
Report per-class IoU, precision, recall, F1 and macro averages. Confusion matrix is row-normalised to percentages.

How Different Models Compare

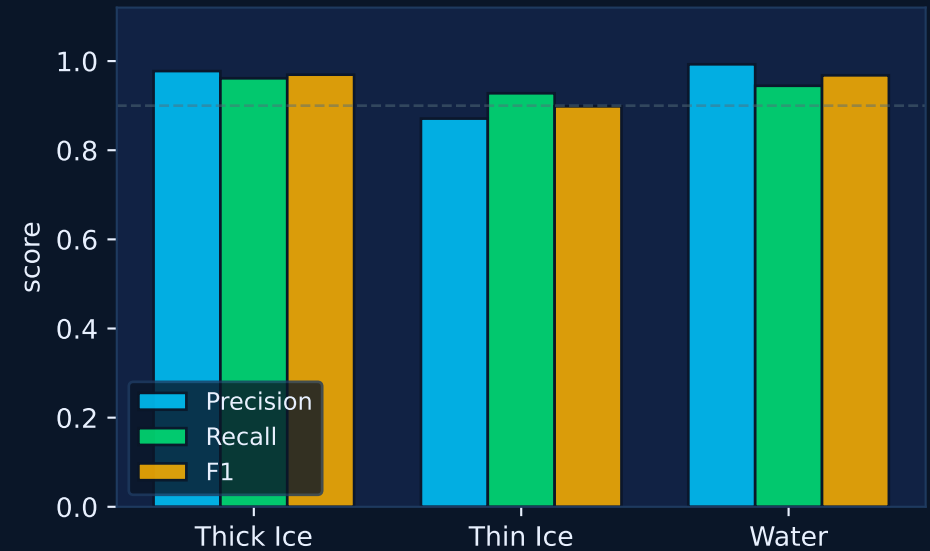
test mIoU — all models



per-class IoU (best model)



precision / recall / F1 (best model)



Best model: Fusion v4 — frozen sweep-winner LSTM + U-Net + SE fusion head

confusion matrix (row %, best model)

	Pred Thick Ice	Pred Thin Ice	Pred Water
Thick Ice	96.14%	3.86%	0.00%
Thin Ice	7.06%	92.78%	0.16%
Water	0.52%	5.01%	94.46%

100

Thick ice — nearly clean

IoU 0.9407 / F1 0.9695
Only ~3% of thick ice pixels get mislabelled. The easiest class.

Thin ice — still the hard one

IoU 0.8157 / F1 0.8985
Gets confused with thick ice most.
The $\alpha=0.45$ weight helps a lot here.

80

60

40

20

0

Open water — texture helps

IoU 0.9382 / F1 0.9681
Recall is 94.5% — the U-Net reads water texture well.

Fusion > single modality

LSTM alone: 0.24 / U-Net: 0.87
Fusion: 0.8982 — about 3 pts above the best single model.

Pretrained LSTM weights paid off

Loading the sweep winner gave +0.6 pp mIoU over random init with no extra training time.

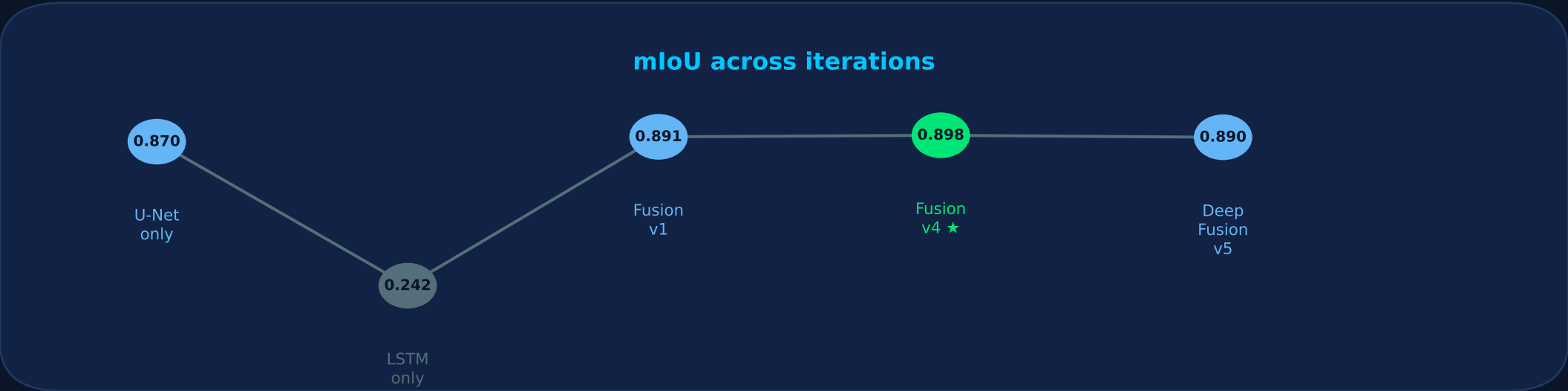
SE block helped a bit

Letting the network re-weight channels after concatenation improved thin ice IoU ~1 pp.

Thin ice is the bottleneck — if you look at where the model struggles, it's almost always that class.

Model	mIoU	pix acc	IoU ice	IoU thin	IoU water
U-Net (image)	0.8704	94.29%	0.9299	0.7683	0.9130
LSTM (CSV only)	0.2420	72.54%	0.7254	0.0000	0.0005
Fusion Deep v1	0.8915	95.12%	0.9396	0.8015	0.9335
Fusion v4 ★	0.8982	95.31%	0.9407	0.8157	0.9382
Deep Fusion v5	0.8896	95.02%	—	—	—

★ best overall (frozen sweep-winner LSTM + U-Net + SE fusion head)



Takeaways

C1

Combining both data sources works — mIoU 0.8982 vs 0.87 for U-Net alone and 0.24 for LSTM alone.

C2

Thin ice is still the weak point (IoU 0.82). The focal loss weighting helps but doesn't fully solve it.

C3

Loading the pre-swept LSTM was worth it. About +0.6 pp mIoU for essentially free.

C4

Tile-grouped splits matter a lot. Random splits inflate scores because tiles overlap spatially.

C5

The SE block is a small but consistent improvement — adds almost no parameters but helps thin ice.

Things worth trying next

F1

Bi-LSTM inside the fusion model — tested standalone and it helped, just haven't wired it in yet.

F2

Test-time augmentation: average a few flipped / rotated predictions to reduce variance a bit.

F3

Stack multiple dates for temporal context — could really help near melt season transitions.

F4

Replace the fixed SE block with a proper attention layer — probably more flexible for edge cases.

F5

Try ATL07 / ATL10 products for more altimetry signal types beyond ATL03.

Stack used

PyTorch 2.x

training + AMP FP16

smp / U-Net

ResNet-18 encoder

ICESat-2 ATL03

along-track altimetry

Focal loss

class imbalance fix

SE attention

channel re-weighting

Fusion v4 — mIoU 0.8982 · pixel acc 95.3% · macro F1 0.9453

thick ice 0.9407 · thin ice 0.8157 · water 0.9382